

JOINT DETECTION OF AI-GENERATED IMAGES AND POST-PROCESSING ALTERATIONS IN REAL-WORLD SCENARIOS

Project Presentation

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Outline

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Problem Statement

- **AI Proliferation:** The widespread use of generative models makes it easy to create photorealistic synthetic images, damaging trust in digital media.
- **The Laboratory Bias:** Existing detectors are trained on original, uncompressed images.
- **Real-World Image Degradation:** Real-world images on social media undergo lossy compression and resizing (**Internet Transfer**), or printing and scanning (**Re-digitisation**).
- **Why Current Detectors Fail:** These post-processing operations erase the delicate high-frequency traces (like camera sensor noise) and others digital artifacts that detectors rely on.

Project Objectives

Task 1: AI Detection (Binary)

Classify if an input image is a genuine photograph or AI-generated (real vs. ai).

Task 2: Transformation Classification (Multi-class)

Identify the post-processing history of the image (original vs. redigitalized vs. internet-transferred).

Unified Multi-Task Framework

Build a single shared backbone model that addresses both questions simultaneously.

Analyze the trade-offs and collaboration between the two tasks using a joint loss function.

State of the Art: AI Detection

- **Spatial CNN Feature Extraction**

CNNs, like **ResNet-50**, are trained with data augmentation (JPEG compression, Gaussian blurring) to classify surface-level textures and color anomalies while generalizing across generative models.

- **Frequency-Domain Detectors**

Utilizing the **2D Discrete Fourier Transform** (DFT) to analyze spectral patterns. Generative networks leave distinct periodic high-frequency artifacts (upsampling fingerprints) that appear as anomalous symmetric peaks in the Fourier magnitude spectrum.

- **The Diffusion Challenge**

Modern **Diffusion Models** (e.g., DDPM) produces realistic spatial textures, bypassing simple convolutional grid detectors. Advanced architectures like **Vision Transformers** (ViTs) capture global contextual dependencies instead of local CNN features.

State of the Art: Robustness

The Challenge

- Real-world images are often affected by **JPEG compression, resizing, and social media processing**.
- These lossy post-processing acts as a low-pass filter, removing **high-frequency traces**, making AI-generated image detection more difficult.

Current Research Solutions

- **Data Augmentation:** A common countermeasure is training models with simulated noise to force them to focus on global structures.
- **Noise-Based Features:** Apply **High-Pass Filtering and Laplacian Pyramids** to extract high-frequency noise residuals rather than semantic content, making features invariant to compression.

State of the Art: Multi-Task & Depth

- **Multi-Task Learning (MLT)**

Learning **AI detection** and **post-processing classification**, i.e. training on related tasks simultaneously acts as an implicit regularizer, forcing the model to learn more general features.

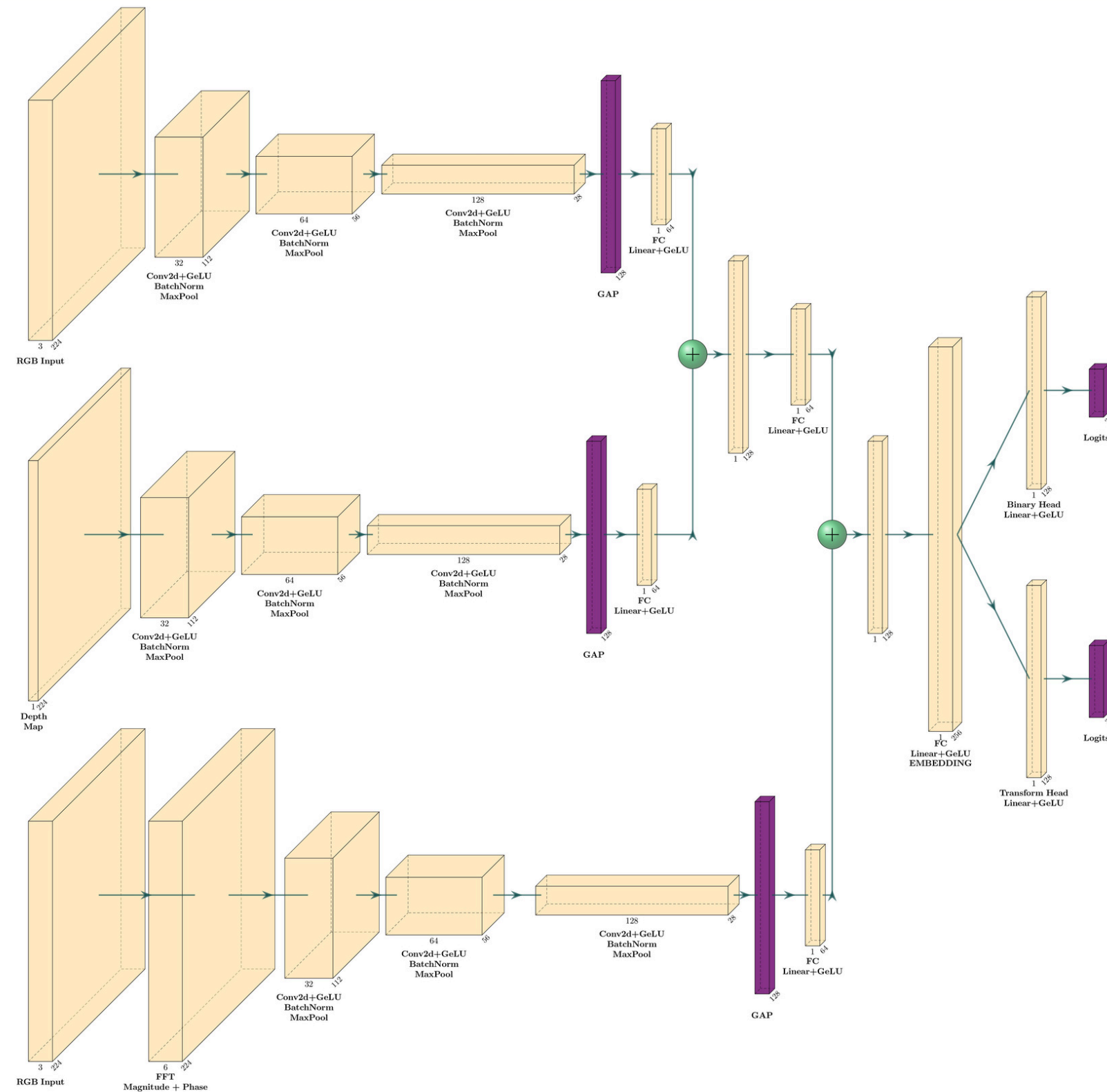
- **Geometric (RGB-D) Forensics**

Combining **RGB images** with **depth information** provides complementary geometric information. AI-generated images may contain inconsistencies in **depth, perspective, and shadow geometry**.

- **Monocular Depth Estimation (MDE)**

Pre-trained models, like **Depth Anything v2**, utilizes a **Teacher-Student training framework with pseudo-labeling** to estimate 3D maps from 2D inputs, without physical depth sensors.

Proposed Method: Overall MultiTaskNet Architecture



Multi-Modal Input Strategy: Fuses spatial textures (**RGB**), estimated 3D geometry (**Depth Map**), and global spectral statistics (**Fourier Spectrum**).

Hierarchical Multi-Stage Fusion:

- **Stage 1: Parallel encoders extract features from each input.** Three separate networks analyze the inputs (RGB, Depth Map, and FFT Magnitude + Phase) at the same time to extract specific characteristics.
- **Stage 2: Intermediate fusion of visual and geometric features.** The global features from the RGB and Depth Map streams are flattened into vectors using Global Average Pooling and summed together to combine visual and spatial information.
- **Stage 3: Late fusion with frequency data.** The combined visual-geometric vector is then summed with the frequency vector (from the FFT stream) to merge all information before sending it to the final classification heads.

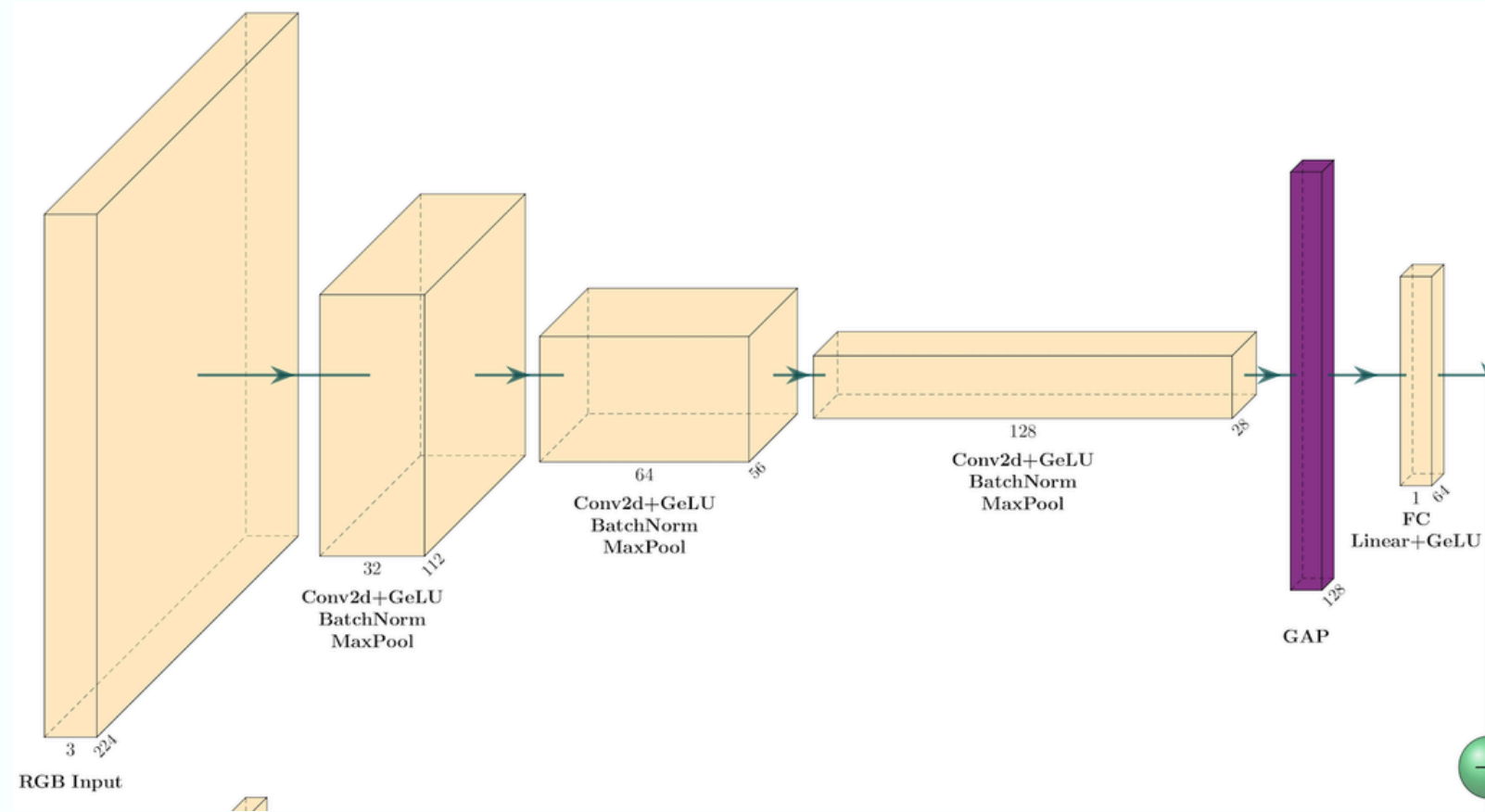
Dual-Head Classification

- **Two independent heads split from the main embedding.** The final shared vector (the 256-D embedding) splits into two parallel Fully Connected (FC) pathways to handle two distinct classification tasks at the same time.
- **Binary Head: AI vs. Real.** This branch focuses on a yes/no decision, classifying whether the image is an original real photograph or an AI-generated one.
- **Transform Head: Tracking the image history.** This parallel branch identifies what happened to the image, choosing between three categories: Original, Re-digitised, or Internet-transferred.

Stage 1: Analyzing Each Input Separately

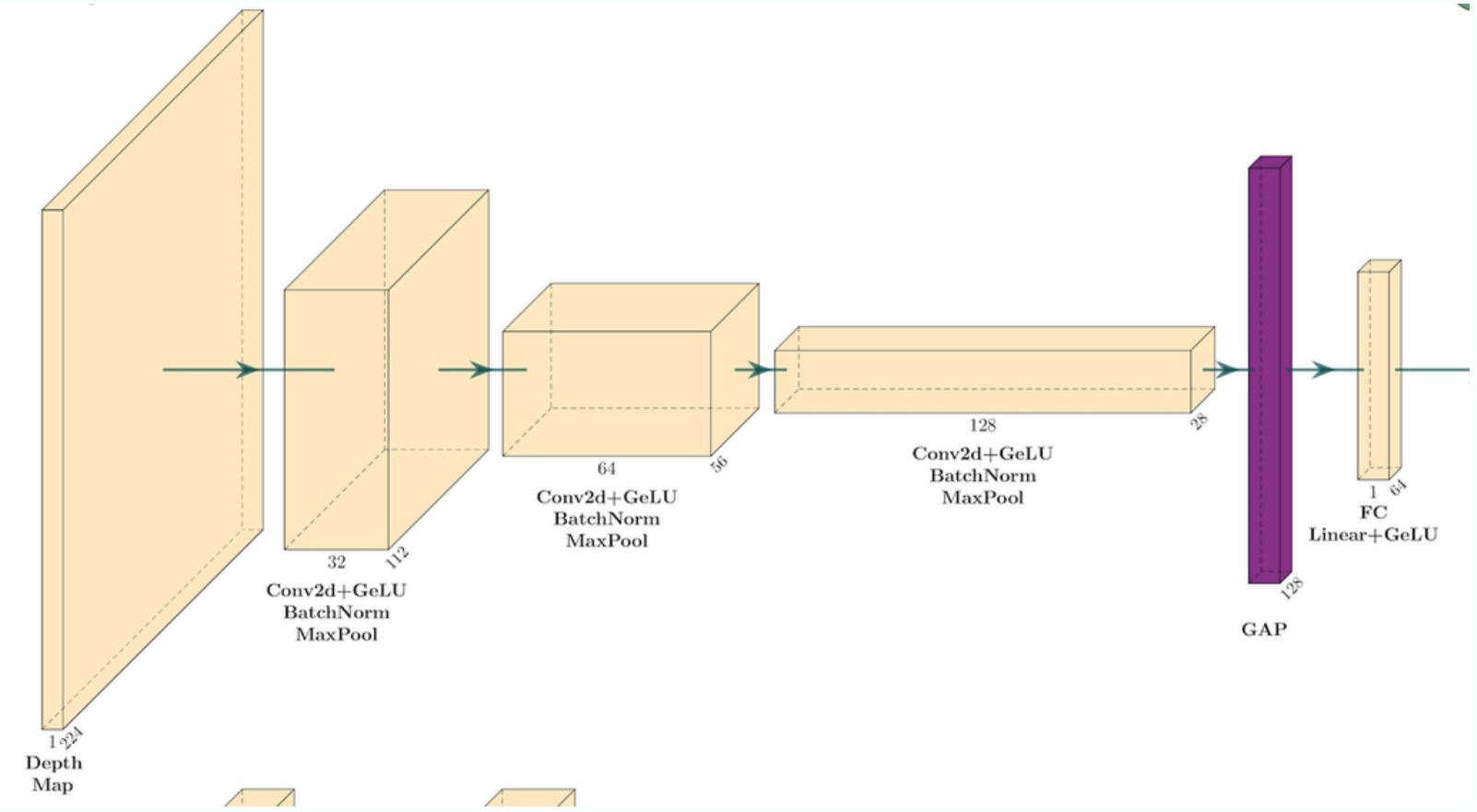
RGBNet (Spatial/Texture Branch)

- Input: $224 \times 224 \times 3$ RGB tensor.
- Encoder: 3 Conv-BatchNorm-GELU-MaxPool blocks + Global Average Pooling.
- Output: 64-D spatial embedding.



DepthNet (3D Geometry Branch)

- Input: $224 \times 224 \times 1$ estimated depth map.
- Encoder: Structurally identical to RGBNet, but processes 1-channel inputs.
- Output: 64-D geometric embedding.



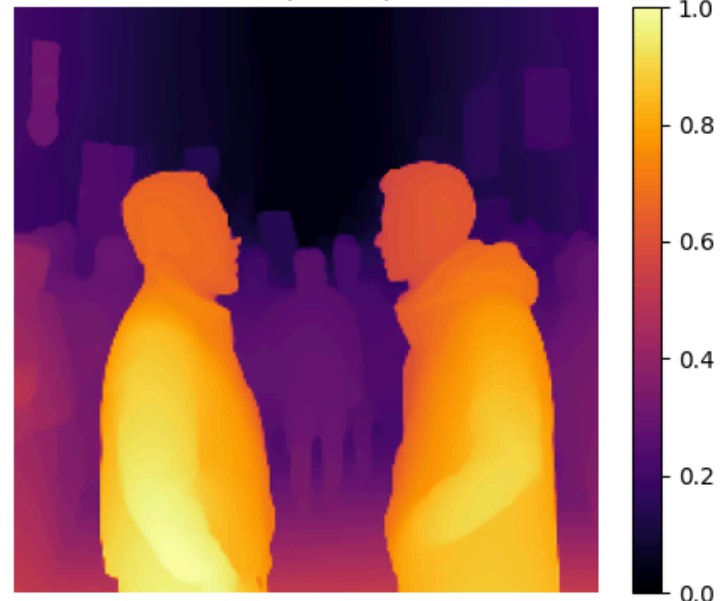
Note: The spatial branches **never share weights** to prevent color contamination in the geometric representation.

Visualizing Estimated Depth Maps

AI Sample



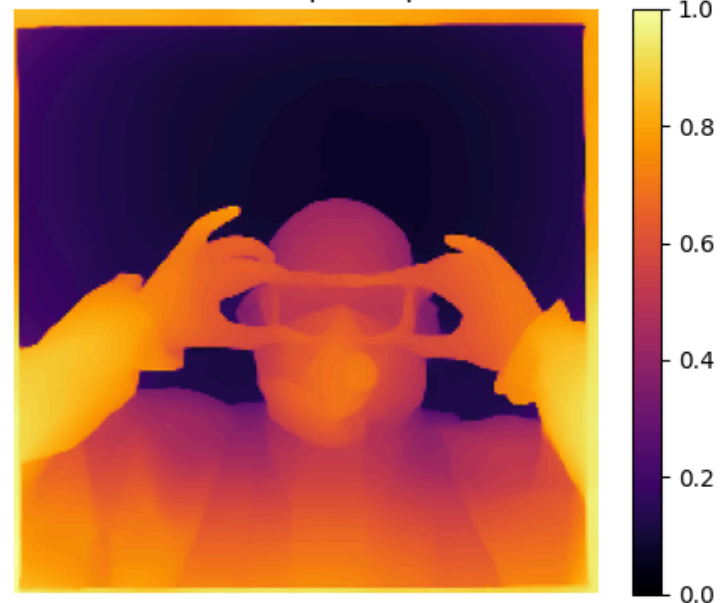
AI Depth Map



REAL Sample



REAL Depth Map



Generative networks struggle to render true 3D spatial properties consistently, yielding detectable volumetric anomalies:

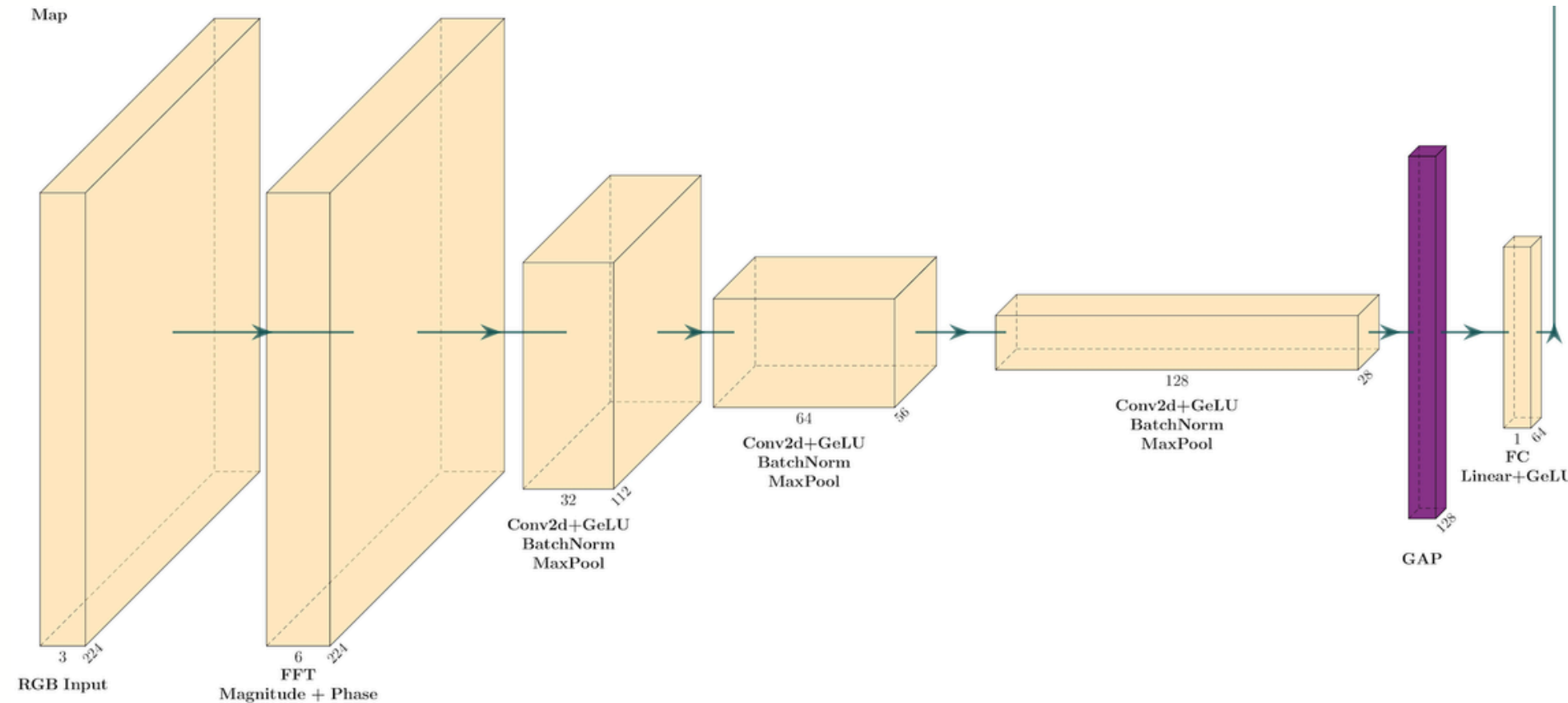
- **Data Source:** Dense depth maps estimated offline using Depth Anything V2 Small model.
- **Normalization:** Every depth map is scaled to a standard range $[0,1]$, where 1.0 represents close foreground and, 0.0 represents the background.
- **Geometric Forensics Utility:** AI-generated images frequently contain perspective flaws and flat background projections around complex boundaries. The estimated depth maps expose these volumetric inconsistencies, which are physically impossible in real camera-captured photographs.

Proposed Method: FourierNet (Frequency Branch)

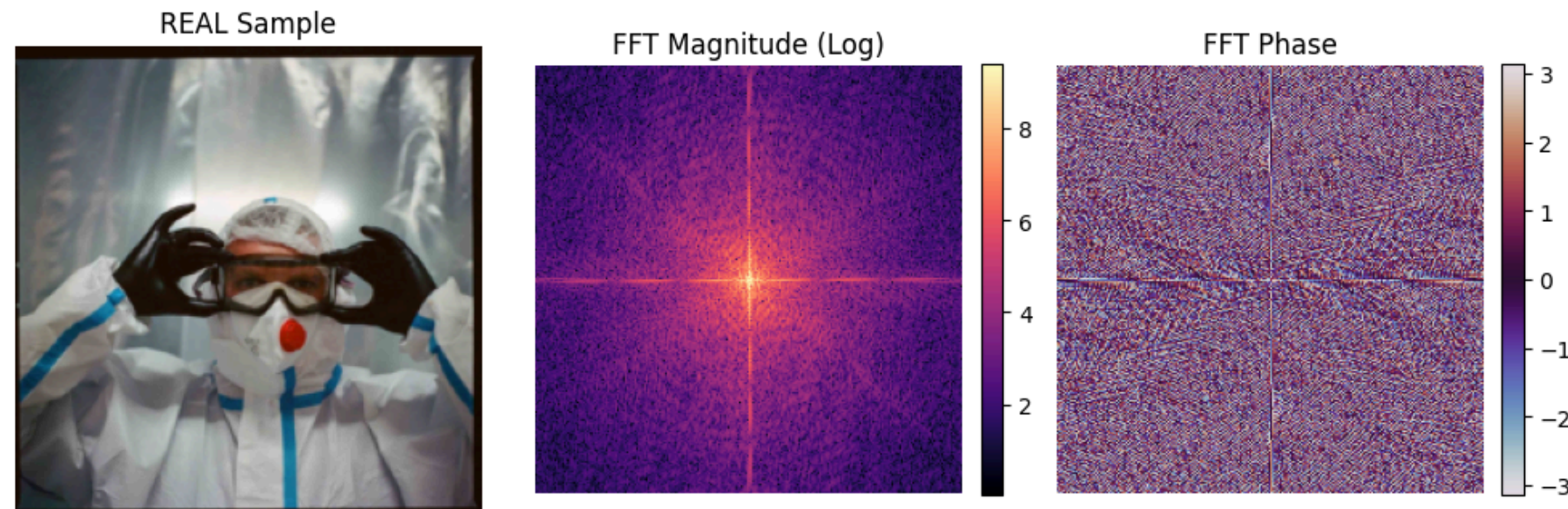
Input Preparation

- *FFT Execution*: Applying 2D Discrete Fast Fourier Transform to the RGB image channels.
- *Logarithmic Compression*: $\log(|X|+1)$ to scale down the dominant low-frequency components, making high-frequency noise patterns visible.
- *Input*: Keep both the magnitude and phase spectra, concatenating them into a $224 \times 224 \times 6$ input tensor.

FourierNet Encoder: 3 Conv-BatchNorm-ReLU-MaxPool blocks, yielding a 64-D spectral embedding.



Visualizing Magnitude and Phase Spectra

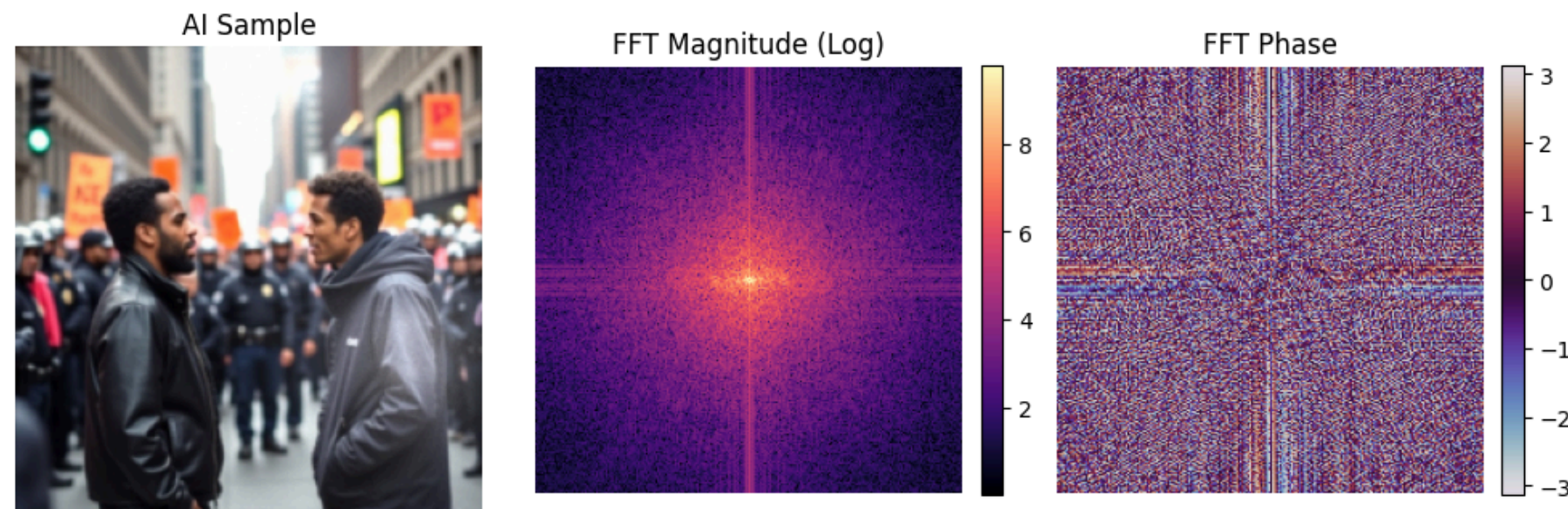


FFT Magnitude

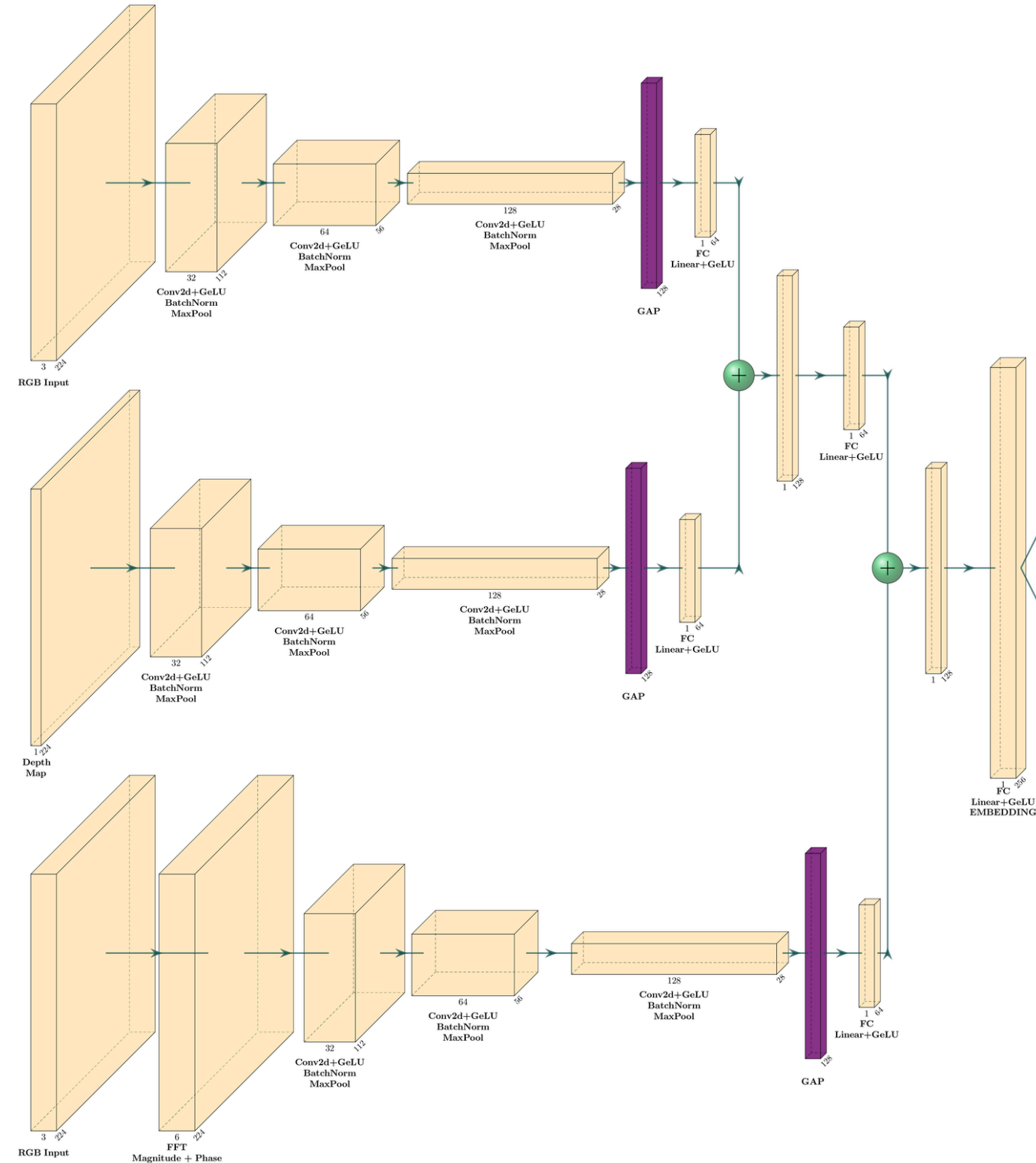
- *What it is:* Measures the strength of each frequency component across the image.
- *Role:* This is where generative models fail. They leave subtle, repetitive patterns that appear as bright, symmetrical spikes or grid-like structures in the magnitude spectrum.

FFT Phase

- *What it is:* Describe the spatial location of these frequencies, capturing where shapes and edges are located.
- *Role:* While appearing as random noise to humans, the phase contains the essential geometric structure (edges and boundaries) of the scene.



Stages 2 & 3: Staged Fusion and Multi-Task Loss



Stage 2: Intermediate Fusion (RGB and Depth)

- *How it works:* The 128-D features from the RGB and Depth streams are reduced to 64-D using individual Fully Connected (FC) layers. These two vectors are then concatenated into a 128-D vector and projected to a 64-D spatial-geometric vector.
- *Reason:* Combining layout (depth) and surface details (RGB) at this point helps the model map structural anomalies directly to texture inconsistencies.

Stage 3: Late Frequency Fusion (Spatial and Frequency)

- *How it works:* Concatenates the spatial-geometric embedding (64-D) with the Fourier embedding (64-D), projecting to the final 256-D unified embedding.
- *Reason:* If we combine the frequency data too early, its global background noise will ruin local details (like edges and shapes) that the convolutional layers are trying to learn.

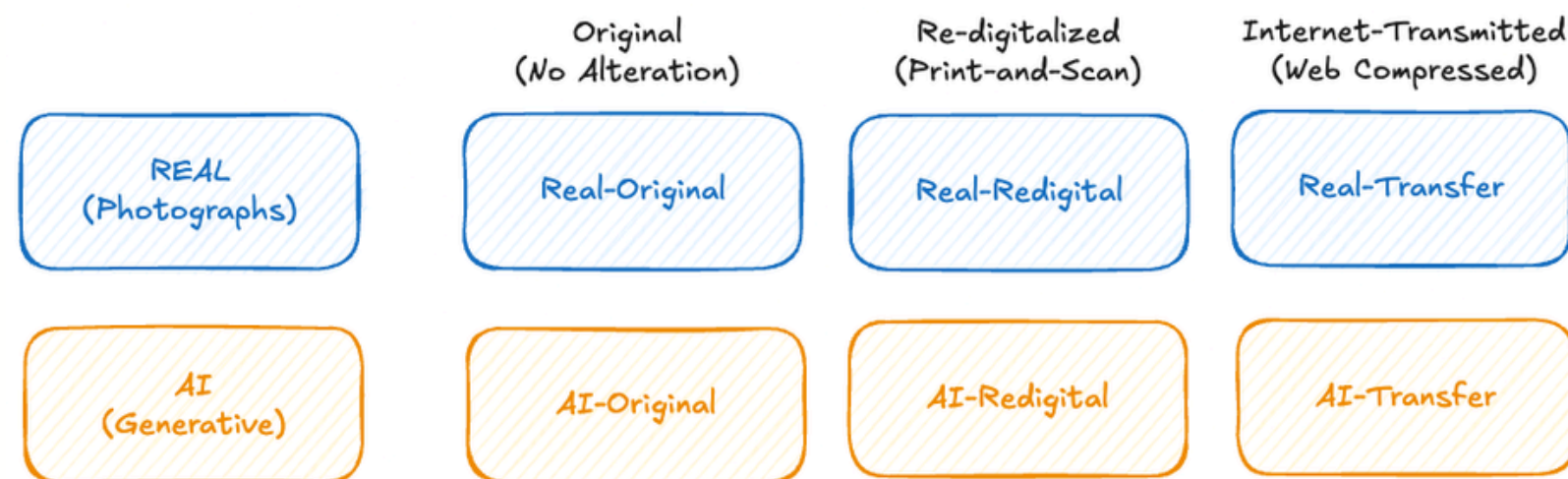
Multi-Task Loss Optimization

The final backbone is trained using a balanced joint loss function:

$$\mathcal{L} = \lambda \cdot \mathcal{L}_{\text{AI}} + (1 - \lambda) \cdot \mathcal{L}_{\text{domain}}$$

This optimization allows the model to simultaneously learn the real/AI boundary ($\mathcal{L}_{\text{AI}}$) and track post-processing changes ($\mathcal{L}_{\text{domain}}$) without one task overtaking the other.

Dataset: The RRDataset and Pre-computed Depth Maps



The RRDataset Structure

- *Task 1 (Image Origin)*: real (captured by physical cameras) vs. ai (synthetically generated).
- *Task 2 (Post-Processing Condition)*:
 - *original*: original, unmodified images.
 - *re-digital*: printed on paper and re-digitized via print-and-scan.
 - *transfer*: uploaded/downloaded online (lossy compression, resizing).

Monocular Depth Dataset

- Pre-computed offline using **Depth Anything v2 (Small)**.
- Cached on disk as single-channel *.pt* tensors of shape
- [1, 224, 224] to avoid high GPU memory consumption during training.

Dataset Statistics & Splits

- Restricted and balanced at **15,000 samples** ($\approx 2,500$ per subgroup) to prevent training bias towards any class.
- Split: **70% Training** (10,500) | **15% Validation** (2,250) | **15% Testing** (2,250), partitioned deterministically using seed 42.

Experimental Setup

Hyperparameter	Value
Optimiser	Adam
Learning rate	1e-4
Batch size	100
Max epochs	50
Early stopping patience	5
Seed	42

Data Pre-processing

- RGB images resized to 224×224 and normalized using standard ImageNet statistics.
- Depth maps loaded from cache and resized to match the RGB spatial resolution (224×224).

Architectural & Efficiency Decisions

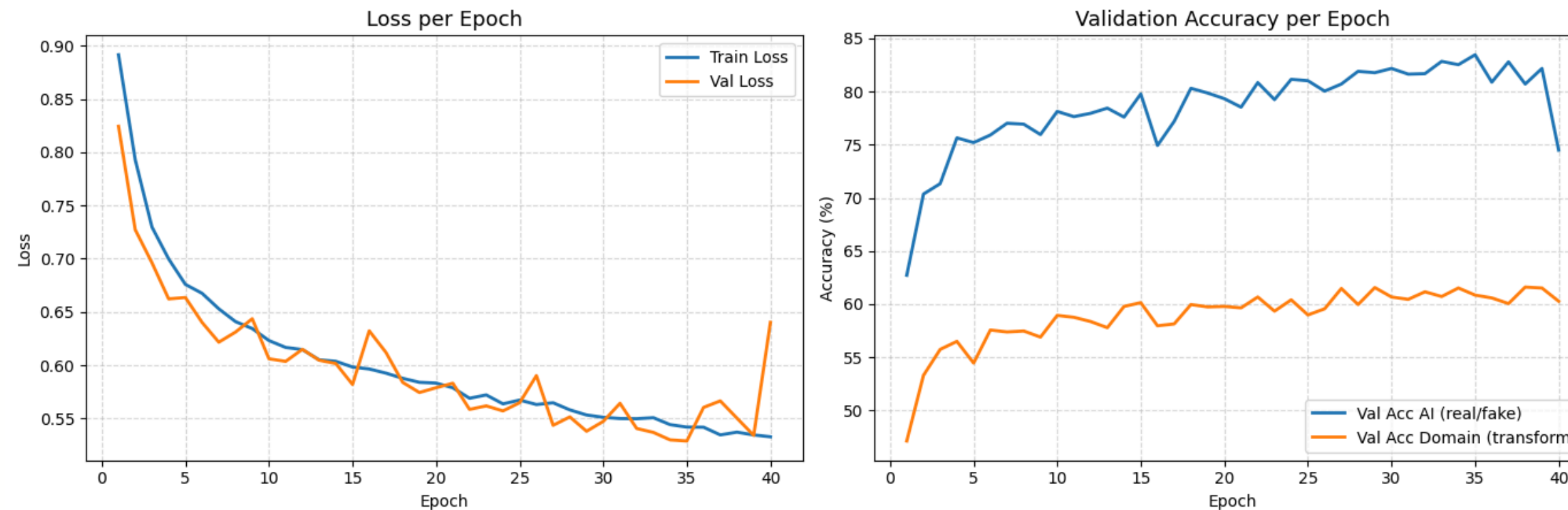
- **Activation Functions:** **GELU** is used in RGBNet, DepthNet, and projection layers to ensure smooth gradients. **ReLU** is used in FourierNet to preserve sharp frequency peaks.
- **Computational Efficiency:** 64-D bottlenecks reduce parameters and calculations for fast inference.
- **Downstream Invariance:** The final 256-D projection ensures classification heads operate on the same tensor size as a standard monotask baseline.

Multi-Task Protocol

- Independent training of 5 models with $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$
- Two unimodal control baselines: AI-only ($\lambda = 1.0$) and Domain-only ($\lambda = 0.0$).

Evaluation: Training Dynamics

Training History - MultiTaskNet ($\lambda = 0.5$)



Loss per Epoch

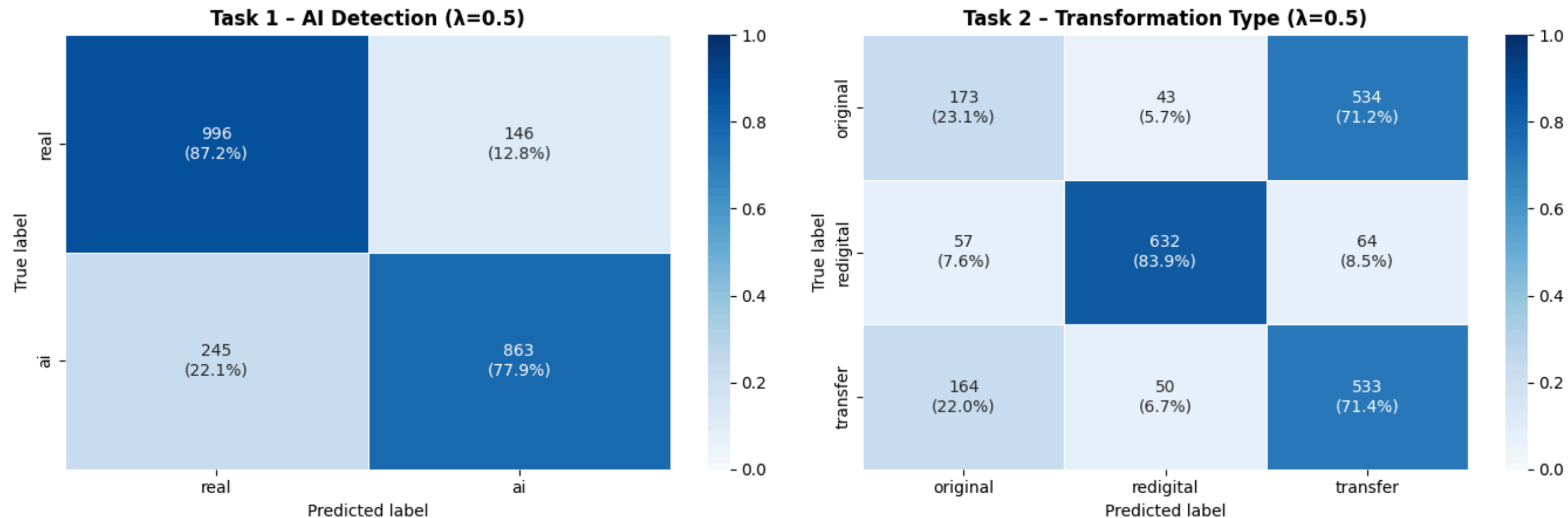
- Training and validation loss decrease steadily and converge closely, showing no signs of overfitting.
- Early stopping successfully terminates training at epoch 40 (saving the best checkpoint at epoch 35).

Validation Accuracy

- Task 1 (AI Detection) accuracy rises quickly, peaking at 83.5% at epoch 35, before dropping at epoch 40 (matching the validation loss spike).
- Task 2 (Domain) accuracy converges smoothly, reaching around 60-61.5%.

Evaluation: Error Analysis via Confusion Matrices

Confusion Matrices - Test Set ($\lambda = 0.5$)



Task 1 (AI Detection)

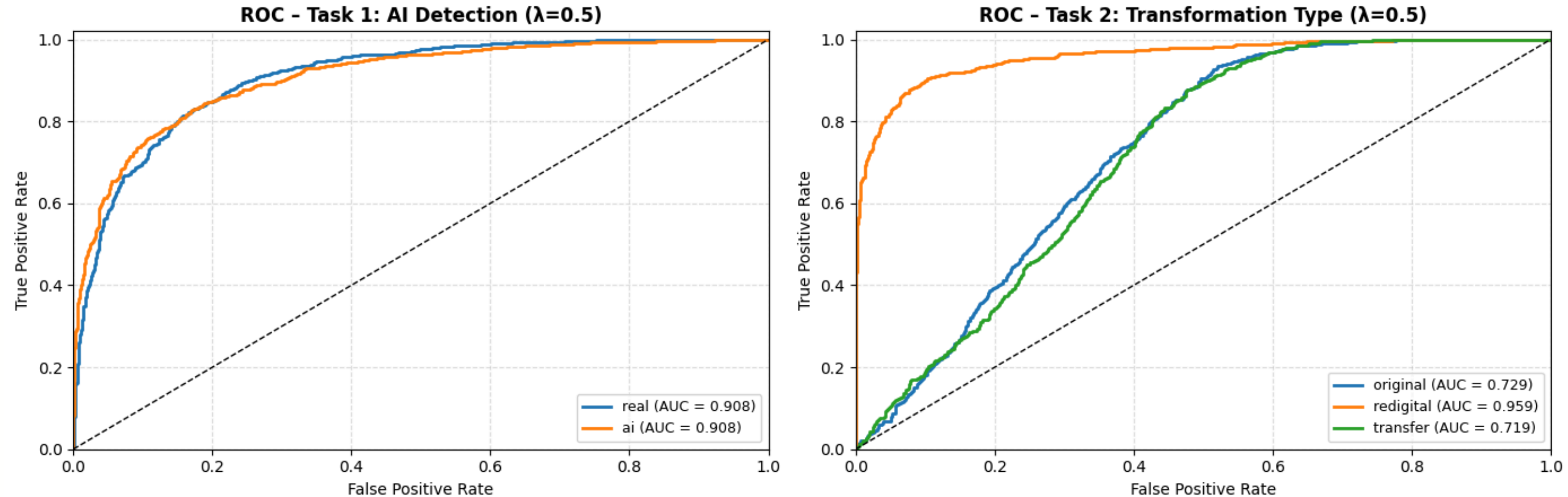
- Overall test accuracy of 82.62%. Real images reach 87.22% recall, while AI images reach 77.89% recall.
- *Observation:* The model is slightly conservative, leaning toward "real" when uncertain (the safest choice).

Task 2 (Transformation Classification)

- *redigital* is highly separable (F1 = 85.52%) due to prominent color shifts and print-and-scan grids.
- *original* and *transfer* are frequently confused with each other. Both are native digital formats sharing similar global frequency patterns.

Evaluation: ROC Curves

ROC Curves - Test Set, One-vs-Rest ($\lambda = 0.5$)



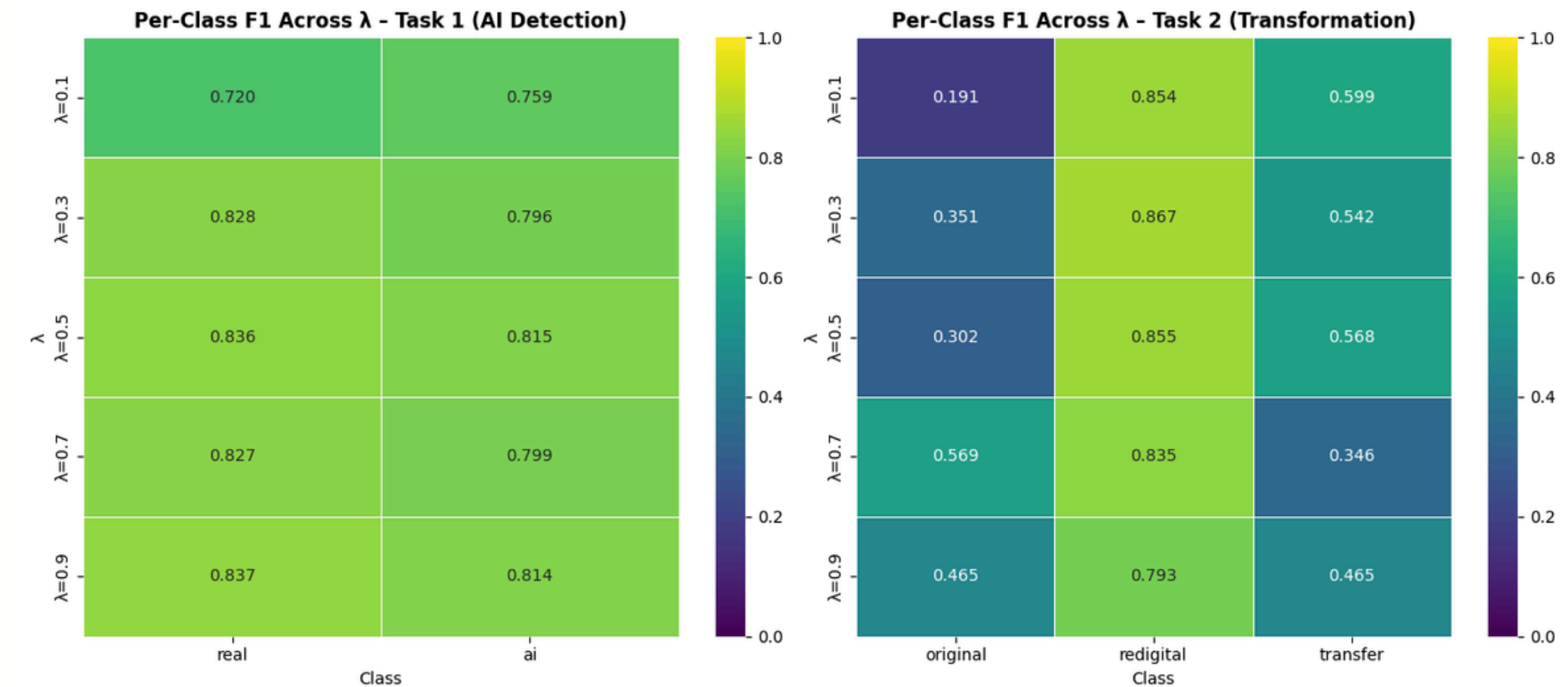
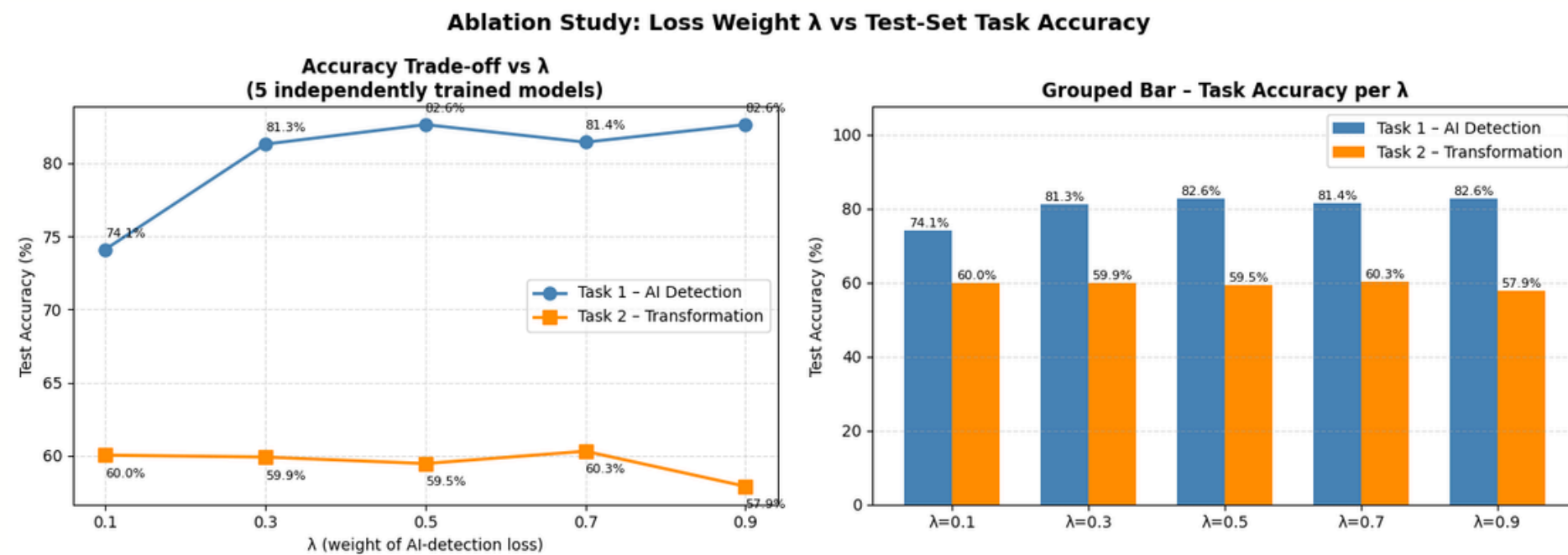
Task 1 (AI Detection)

- Achieves an excellent AUC of 0.908 for both classes, showing high separability.

Task 2 (Transformation Classification)

- *redigital* has an outstanding AUC of 0.959, confirming it is easily isolated by our spectral features.
- *original* (AUC = 0.729) and *transfer* (AUC = 0.719) reflect the boundary confusion, but still show solid classification capability far above random guess (0.5).

Evaluation: Task Loss Weighting Trade-off



Task 1 Accuracy

Rises from 74.09% ($\lambda=0.1$) and plateaus at 82.62% ($\lambda=0.5$).
Further increases in λ yield no gains.

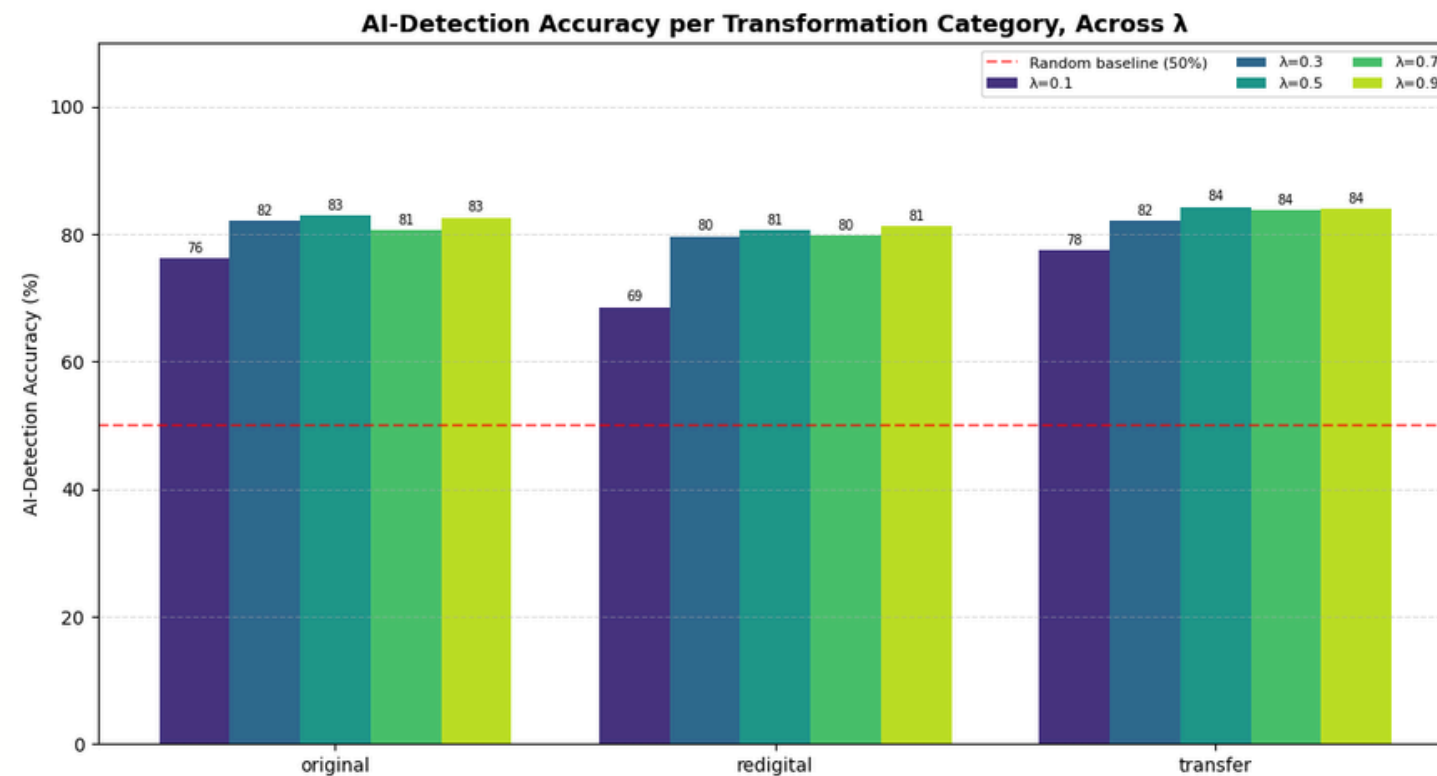
Task 2 Accuracy

Remains stable between 57.91% and 60.31% across all values.

F1 Heatmaps (Class-Specific Trade-off)

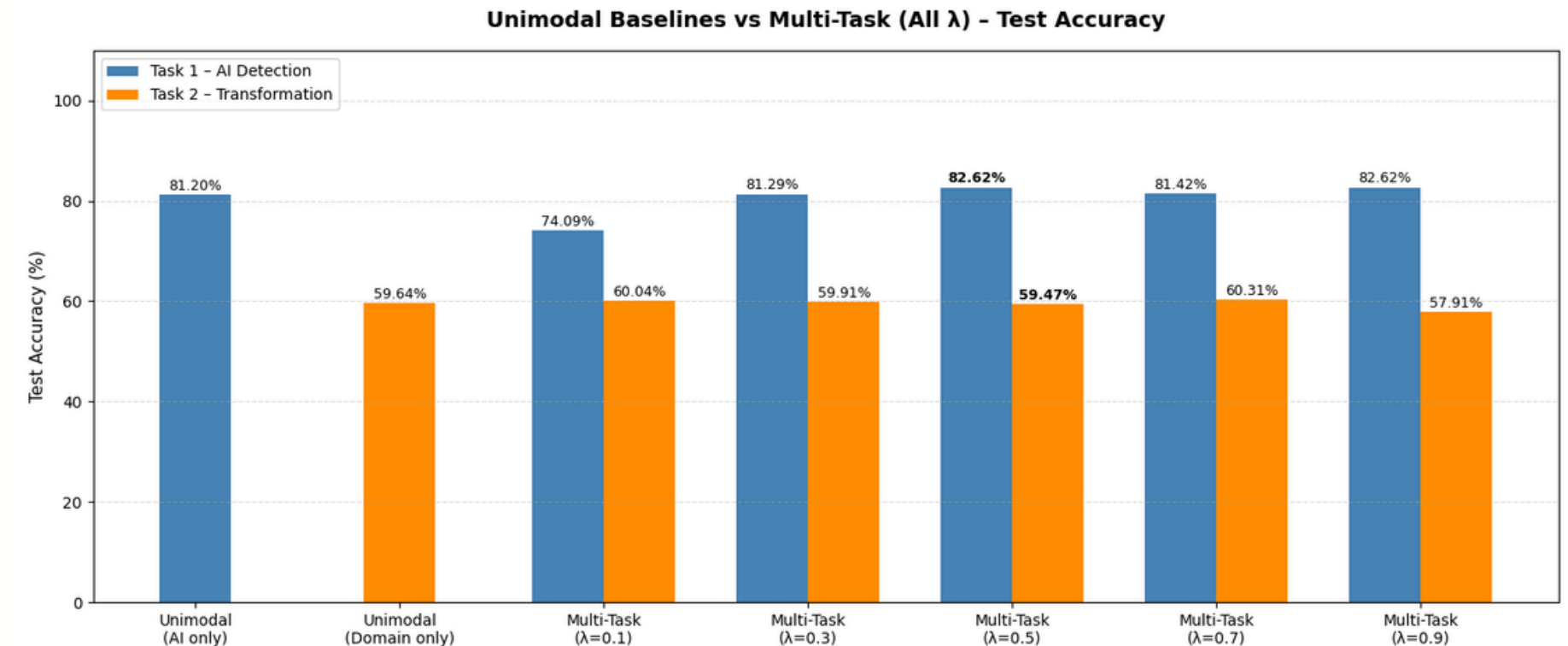
- *original* and *transfer* compete for representation capacity.
- At $\lambda=0.1$, transfer peaks (F1 = 0.599) but original collapses (F1 = 0.191).
- At $\lambda=0.7$, original peaks (F1 = 0.569) but transfer drops (F1 = 0.346).
- $\lambda=0.5$ provides the most balanced compromise without causing either class to collapse.

Evaluation: Robustness Under Degradation & Baseline Comparison



AI Detection Accuracy by Domain ($\lambda=0.5$)

- *Original*: 82.93% | *Internet-transferred*: 84.34% | *Re-digitalised*: 80.61%.
- AI detection drops by **only 2.3%** on re-digitized images, proving that DepthNet (3D geometry) and FourierNet (phase alignments) successfully compensate for lost spatial textures.



Unimodal Baselines vs. Multi-Task

- *AI-only baseline* ($\lambda=1.0$): 81.2% Task 1 accuracy (achieves results equivalent to the level of randomness in Task 2).
- *Domain-only baseline* ($\lambda=0.0$): 59.6% Task 2 accuracy (achieves results equivalent to the level of randomness in Task 1).
- *Multi-Task model* ($\lambda=0.5$): **82.6%** on Task 1 (outperforming the unimodal!) and **59.5%** on Task 2.

Conclusions

Multi-Modal Synergy

Combining spatial textures (RGB), estimated geometry (Depth), and global frequencies (Fourier) protects the network from single-modality failures under post-processing.

Staged Fusion Design

Fusing aligned spatial features early and non-spatial frequency features late successfully prevents spectral statistics from corrupting local CNN filters.

Multi-Task Regularization and Efficiency

Joint multi-task training forces the shared backbone to learn representations that are invariant to compression and scan distortions.

A single joint model matches or exceeds the accuracy of two unimodal specialists, saving half the parameters and inference time.

Conclusions: Limitations & Future Work

Limitations

- *Dataset Scale*: The 15,000 sample boundary limits diversity across different real-world scenarios.
- *Depth Map Precision*: Depth maps estimated by Depth Anything V2 Small are only approximations and they can become noisy and inaccurate in complex scenes.

Future Directions

- *Robustness under Image Degradation*: Implement an advanced data augmentation pipeline that simulates common real-world artifacts, such as JPEG compression, sensor noise, and blur. This would test and improve the model's reliability when processing lower-quality inputs.
- *State-of-the-Art Architectures*: Replacing the existing CNN-based feature extraction branches with more powerful models like Vision Transformers (ViTs). These models could extract richer, more global semantic representations, potentially improving overall performance.

References

[1] Sapienza University of Rome Computer Vision Lecture Notes, A.Y. 2025-2026.

[2] **Li, C., et al.** "*Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios.*" Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.

[3] **Shao, R., Wu, T., and Liu, Z.** "*Detecting and grounding multi-modal media manipulation.*" Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.